

§11.4.29 CamelCase → snake_case Upstream Repo Rename Plan

Date: 2026-06-08 **Status:** PLAN ONLY — uncommitted, operator/main-agent executes deliberately. **Mandate:** HelixConstitution §11.4.29 (own-org repos use snake_case names). **Constraints honored:** §6.T.3 (no force-push, per-operation operator approval), §6.W (GitHub + GitLab parity only), §6.U (no sudo/su).

0. Executive summary — the actual problem is smaller than it looks

The 5 dep submodules are already **mounted at snake_case paths** on disk (submodules/doc_processor, ...) and their Go **module paths are already lowercase logical names** (digital.vasic.docprocessor, ...) that do NOT embed the repo name. The CamelCase only survives in three places:

1. The **remote repo NAMES** on GitHub + GitLab (vasic-digital/DocProcessor, ...).
2. The **.gitmodules url= entries** in the Lava parent (PascalCase).
3. A handful of **replace ... => ../<PascalCasePath>** directives — and the load-bearing one (helixqa upstream main go.mod) is exactly why the helixqa pin must track lava-pin/2026-06-08-android-executor instead of main. The lava-pin branch already uses ../doc_processor; only upstream main uses ../DocProcessor.

Because the Go module identifiers do not change, **renaming the remote repos does NOT cascade into Go import paths or go.sum checksums**. The rename is almost entirely a metadata operation (repo name + .gitmodules url + submodule remote URLs).

Bonus bug found (fix in the same effort): submodules/llm_orchestrator/go.mod:18 is replace digital.vasic.llmprovider => ../LLMProvider but the disk path is llm_provider. This works only by macOS case-insensitivity and is broken on a case-sensitive volume. It must become => ../llm_provider.

(a) Mapping table

Verified via `git -C submodules/<name> remote -v, gh repo view, glab repo view`. snake_case targets confirmed to NOT yet exist on either provider.

submodule path	current GitHub repo	current GitLab repo	proposed snake_case
submodules/doc_processor	vasic-digital/ DocProcessor (origin); also HelixDevelopment/ DocProcessor exists	vasic-digital/ DocProcessor	doc_processor
submodules/llm_orchestrator	vasic-digital/ LLMOrchestrator (origin fetch); also HelixDevelopment/ LLMOrchestrator exists	vasic-digital/ LLMOrchestrator	llm_orchestrator
submodules/llm_provider	vasic-digital/ LLMProvider (origin); also HelixDevelopment/ LLMProvider exists	vasic-digital/ LLMProvider	llm_provider
submodules/llms_verifier	vasic-digital/ LLMsVerifier (origin)	vasic-digital/ LLMsVerifier	llms_verifier
submodules/vision_engine	vasic-digital/ VisionEngine (origin fetch); also HelixDevelopment/ VisionEngine exists	vasic-digital/ VisionEngine	vision_engine
panoptic (other own-org CamelCase)	vasic-digital/ Panoptic	vasic-digital/ panoptic (already lowercase on GitLab)	panoptic

Already snake_case / lowercase (NO rename needed): all 16 core submodules (auth, cache, challenges, concurrency, config, containers, database, discovery, http3, mdns, middleware, observability, ratelimiter, recovery, security, tracker_sdk). HelixConstitution (HelixDevelopment/
HelixConstitution) is constitution-owned — out of scope, not subject to §11.4.29 lowercasing for Lava per §6.AD.1. helixqa (HelixDevelopment/helixqa on GitHub, helixdevelopment1/HelixQA on GitLab) is already lowercase on GitHub; the GitLab HelixQA casing is a HelixDevelopment-org concern.

Remote-inconsistency flags (MUST be reconciled, see §c step 0)

The 5 dep submodules have **divergent, non-§6.W-compliant remote sets** today:

- doc_processor: clean (origin=GitHub vasic-digital, gitlab=GitLab vasic-digital). 
- llm_orchestrator: origin fetch=vasic-digital, origin PUSH=HelixDevelopment; extra github+upstream to HelixDevelopment. × mixed-org.
- llm_provider: clean. 

- `llms_verifier`: extra github+upstream duplicating origin; origin has a stray second PUSH url to GitLab. ✗ malformed.
- `vision_engine`: origin fetch=vasic-digital, origin PUSH=HelixDevelopment; extra github+upstream to HelixDevelopment. ✗ mixed-org.

These predate the rename and must be normalized to the §6.W 2-remote model (origin=GitHub, gitlab=GitLab, both vasic-digital) BEFORE renaming, so the rename targets are unambiguous.

(b) Blast-radius inventory (file:line evidence)

B1. Parent-tracked files (the only files THIS repo's commits touch)

- **.gitmodules** — 5 dep url lines + panoptic:
 - `url = git@github.com:vasic-digital/DocProcessor.git` (doc_processor block)
 - `url = git@github.com:vasic-digital/LLMOrchestrator.git` (llm_orchestrator block)
 - `url = git@github.com:vasic-digital/LLMProvider.git` (llm_provider block)
 - `url = git@github.com:vasic-digital/LLMsVerifier.git` (llms_verifier block)
 - `url = git@github.com:vasic-digital/VisionEngine.git` (vision_engine block)
 - `url = git@github.com:vasic-digital/Panoptic.git` (panoptic block)
- **docs/CONTINUATION.md** — descriptive PascalCase references (§6.S doc, update text):
 - :17 — replace ../DocProcessor narrative
 - :390 — LLMOrchestrator narrative
 - :430–434 — pin-index table rows naming vasic-digital/DocProcessor, .../LLMOrchestrator, .../LLMProvider, .../LLMsVerifier, .../VisionEngine
- **constitution/submodules-catalogue.md** — constitution-submodule-owned; references the names. NOT a Lava-parent commit (lives in the constitution/ submodule git-dir). Flag to operator: update upstream in HelixConstitution if it pins repo names.

B2. go.mod / go.sum references (verified by grep)

```
submodules/helixqa/go.mod:127    digital.vasic.docprocessor
=> ../doc_processor    (lava-pin: already snake_case ✅)
submodules/helixqa/go.mod:128    digital.vasic.llmorchestrator
=> ../llm_orchestrator (✅)
submodules/helixqa/go.mod:129    digital.vasic.llmprovider
=> ../llm_provider     (✅)
submodules/helixqa/go.mod:130    digital.vasic.llmsverifier
=> ../llms_verifier/llm-verifier (✅)
submodules/helixqa/go.mod:132    digital.vasic.visionengine
=> ../vision_engine    (✅)
```

```
submodules/llm_orchestrator/go.mod:18  replace
digital.vasic.llmprovider => ../LLMProvider  x BROKEN PATH (disk
is llm_provider)
```

- helixqa **upstream main** go.mod (git show origin/main:go.mod):126-132 uses ../DocProcessor, ../LLMOrchestrator, ../LLMProvider, ../LLMsVerifier/llm-verifier, ../VisionEngine — PascalCase. This is the divergence the lava-pin branch fixes. Once the repos rename, the path-side fix can be merged upstream so main works on case-sensitive volumes and the pin can return to tracking main.
- **NO go.sum churn** — module paths (digital.vasic.*) are unchanged by a repo rename; checksums unaffected.
- lava-api-go/go.mod references only digital.vasic.helixqa => ../submodules/helixqa — unaffected.

B3. Go import paths in source

None. Modules use logical digital.vasic.* identifiers, not github.com/owner/Repo paths. grep for vasic-digital/<Pascal> and HelixDevelopment/<Pascal> in *.go returned zero matches in tracked source.

B4. Submodule-internal files (owned by the submodule repos, NOT Lava parent)

These change inside each submodule's own repo, in that repo's own commits — NOT in a Lava-parent commit. Listed so the operator updates them upstream:

- submodules/doc_processor/{README.md, USER_GUIDE.md, helix-deps.yaml, upstreams/VasicDigitalGitHub.sh, upstreams/VasicDigitalGitLab.sh, upstreams/GitHub.sh}
- submodules/llm_orchestrator/{helix-deps.yaml, CONTRIBUTING.md, USER_GUIDE.md, docs/test-coverage.md, upstreams/*.sh}
- submodules/llm_provider/{helix-deps.yaml, upstreams/*.sh}
- submodules/llms_verifier/{README.md, CLEANUP_CHECKLIST.md, *_PLAN.md, SCORING_SYSTEM_DOCUMENTATION.md, ...}
- submodules/vision_engine/* (analogous)
- Each helix-deps.yaml header comment names the repo (e.g. doc_processor's says "vasic-digital/DocProcessor") — cosmetic; update for §11.4.31 honesty.
- The upstreams/*.sh scripts hardcode the PascalCase repo in push URLs — **functionally load-bearing inside the submodule**; must be updated or they push to the now-redirected old name.

B5. CI / scripts in the parent

grep over scripts/, Upstreams/, tools/ for the 5 PascalCase names: **zero matches**. The parent's mirror scripts do not name these repos. 

(c) Exact ordered command sequence

Run from repo root /Volumes/T7/Projects/Lava. Read-only verification steps are safe; mutating steps (gh repo rename, glab, git remote set-url, edits, git submodule sync) are the deliberate execution.

Step 0 — Pre-flight backup + remote normalization (REQUIRED first)

```
# 0a. $9 hardlinked .git backup before any metadata mutation
cp -al .git ".git-backup-pre-snakecase-$(date +%Y%m%d-%H%M%S)/
repo.git.mirror"

# 0b. Normalize the divergent submodule remotes to the $6.W 2-
remote model.
# (Do this for llm_orchestrator, llms_verifier, vision_engine
which have
# mixed-org / malformed remotes.)
for s in llm_orchestrator vision_engine; do
    git -C submodules/$s remote remove upstream 2>/dev/null || true
    git -C submodules/$s remote remove github 2>/dev/null || true
    git -C submodules/$s remote set-url origin git@github.com:vasic-
digital/${
        case $s in llm_orchestrator) echo LLMOrchestrator;;
        vision_engine) echo VisionEngine;; esac
    }.git
    git -C submodules/$s remote set-url --push origin
    git@github.com:vasic-digital/${
        case $s in llm_orchestrator) echo LLMOrchestrator;;
        vision_engine) echo VisionEngine;; esac
    }.git
done
git -C submodules/llms_verifier remote remove upstream 2>/dev/
null || true
git -C submodules/llms_verifier remote remove github 2>/dev/
null || true
git -C submodules/llms_verifier remote set-url --add --
push origin git@github.com:vasic-digital/LLMsVerifier.git
# then drop the stray gitlab push
# Verify each ends up with exactly: origin=GitHub vasic-digital,
gitlab=GitLab vasic-digital
for s
    in doc_processor llm_orchestrator llm_provider
    llms_verifier vision_engine; do
    echo "== $s =="; git -C submodules/$s remote -v
done
```

OPERATOR APPROVAL GATE 1 — confirm the normalized remote sets before any rename.

Step 1 — Rename the remote repos (GitHub + GitLab), one repo at a time

GitHub `gh repo rename` and GitLab `path-rename` both install permanent HTTP/SSH **redirects** from the old name, so existing clones keep fetching/pushing until their remote URL is updated. (GitHub: documented redirect. GitLab: documented "Repository path redirect" on project path change — verify the redirect is live with the fetch in Step 3 before deleting any old reference.)

doc_processor

```
gh repo rename doc_processor --repo vasic-digital/DocProcessor
glab repo edit vasic-digital/DocProcessor --path
doc_processor      # path-rename on GitLab
```

llm_orchestrator

```
gh repo rename llm_orchestrator --repo vasic-digital/
LLMOrchestrator
glab repo edit vasic-digital/LLMOrchestrator --path
llm_orchestrator
```

llm_provider

```
gh repo rename llm_provider --repo vasic-digital/LLMProvider
glab repo edit vasic-digital/LLMProvider --path llm_provider
```

llms_verifier

```
gh repo rename llms_verifier --repo vasic-digital/LLMsVerifier
glab repo edit vasic-digital/LLMsVerifier --path llms_verifier
```

vision_engine

```
gh repo rename vision_engine --repo vasic-digital/VisionEngine
glab repo edit vasic-digital/VisionEngine --path vision_engine
```

panoptic (GitHub PascalCase → lowercase; GitLab already lowercase)

```
gh repo rename panoptic --repo vasic-digital/Panoptic
# glab: GitLab already vasic-digital/panoptic – no path change
needed; verify only.
```

NOTE: `glab repo edit --path` is the GitLab path-rename verb. If the installed `glab` lacks `--path`, fall back to the GitLab REST PUT: `glab api --method PUT projects/vasic-digital%2FDocProcessor -f path=doc_processor`. Do NOT change the GitLab **display name** (name) — only the **path**.

OPERATOR APPROVAL GATE 2 — the `HelixDevelopment-org` duplicates (`HelixDevelopment/DocProcessor`, `.../LLMOrchestrator`, `.../LLMProvider`, `.../VisionEngine`) ALSO exist. Decide per-repo whether they are stale and should be renamed/archived too, or left alone. They are NOT referenced by Lava's current origin set after Step 0, so they are non-blocking — but §11.4.29 applies to them if they are live own-org repos.

Step 2 — Point each submodule's local remotes at the new names

```
git -C submodules/doc_processor remote set-url origin
    git@github.com:vasic-digital/doc_processor.git
git -C submodules/doc_processor remote set-url gitlab
    git@gitlab.com:vasic-digital/doc_processor.git
git -C submodules/llm_orchestrator remote set-url origin
    git@github.com:vasic-digital/llm_orchestrator.git
git -C submodules/llm_orchestrator remote set-url gitlab
    git@gitlab.com:vasic-digital/llm_orchestrator.git
git -C submodules/llm_provider remote set-url origin
    git@github.com:vasic-digital/llm_provider.git
git -C submodules/llm_provider remote set-url gitlab
    git@gitlab.com:vasic-digital/llm_provider.git
git -C submodules/llms_verifier remote set-url origin
    git@github.com:vasic-digital/llms_verifier.git
git -C submodules/llms_verifier remote set-url gitlab
    git@gitlab.com:vasic-digital/llms_verifier.git
git -C submodules/vision_engine remote set-url origin
    git@github.com:vasic-digital/vision_engine.git
git -C submodules/vision_engine remote set-url gitlab
    git@gitlab.com:vasic-digital/vision_engine.git
git -C panoptic remote set-url origin
    git@github.com:vasic-digital/panoptic.git
git -C panoptic remote set-url github
    git@github.com:vasic-digital/panoptic.git
git -C panoptic remote set-url gitlab
    git@gitlab.com:vasic-digital/panoptic.git
```

Step 3 — Update parent .gitmodules + sync

```
# Edit .gitmodules url= lines (PascalCase → snake_case):
#   vasic-digital/DocProcessor.git   -> vasic-digital/
#       doc_processor.git
#   vasic-digital/LLMOrchestrator.git -> vasic-digital/
#       llm_orchestrator.git
#   vasic-digital/LLMProvider.git     -> vasic-digital/
#       llm_provider.git
#   vasic-digital/LLMsVerifier.git     -> vasic-digital/
#       llms_verifier.git
#   vasic-digital/VisionEngine.git     -> vasic-digital/
#       vision_engine.git
#   vasic-digital/Panoptic.git         -> vasic-digital/
#       panoptic.git
# (use an editor / Edit tool; paths in [submodule "..."] headers
#   stay as-is -
#   they already match the snake_case on-disk paths.)

git submodule sync --recursive
# Verify the redirect + new URL both resolve:
for s
    in doc_processor llm_orchestrator llm_provider
    llms_verifier vision_engine; do
```

```
echo "== $s =="; git -C submodules/$s ls-remote origin -h 2>&1  
| head -1  
done
```

Step 4 — Fix the broken/stale replace path (independent bug)

```
# submodules/llm_orchestrator/go.mod:18  
# replace digital.vasic.llmprovider => ../LLMProvider  
# ->  
# replace digital.vasic.llmprovider => ../llm_provider  
# Commit inside the llm_orchestrator repo, push to both mirrors.
```

Step 5 — Submodule-internal upstream/doc updates (each in its own repo)

For each of the 5 dep repos (and panoptic), inside the submodule:

- update upstreams/*.sh push URLs to the new lowercase repo name,
- update helix-deps.yaml header comment + any ssh_url/org field,
- update README/docs PascalCase repo references,
- commit + push to origin (GitHub) and gitlab (GitLab) — §6.W parity.

Step 6 — helixqa upstream realignment (optional but closes the root cause)

Once the repos are snake_case, the path-side fix (../doc_processor etc.) that lives on lava-pin/2026-06-08-android-executor can be merged into helixqa main (a normal PR, no force-push). After that the Lava helixqa pin can return to tracking main and the §11.4.29 OWED item + the lava-pin workaround both close. Update docs/CONTINUATION.md §0/§3 accordingly.

Step 7 — Parent commit (Lava)

In ONE Lava-parent commit (branch off master first per commit policy):

- .gitmodules url updates,
- the 6 submodule gitlink pins (if any submodule SHA advanced from Step 4/5),
- docs/CONTINUATION.md updates (pin table rows, §11.4.29 OWED → DONE, helixqa pin note),
- then push to GitHub + GitLab, verify §6.C 2-mirror convergence.

(d) Risk & sequencing assessment

Independent vs cascading

- **The repo renames are mutually independent** — no module-path changes, so renaming doc_processor does not ripple into llm_orchestrator etc. They can be done in any order.

- **The only intra-Go cascade** is `llm_orchestrator` → `llm_provider` via the `replace ../LLMProvider` line — but that is a *path* fix (Step 4), not a *module-id* change, and is already correct on the helixqa side. Fix it regardless; it is currently broken on case-sensitive volumes.
- **Sequencing that matters:** Step 0 (remote normalization) **MUST** precede Step 1 (rename) so the rename targets are unambiguous; Step 1 **MUST** precede Steps 2–3 (local URL updates rely on the new names existing + redirecting). Within Step 1, GitHub-then-GitLab order is interchangeable.

Redirect safety

- GitHub `gh repo rename` → permanent redirect; old clones keep working. LOW risk.
- GitLab `path-rename` → documented path redirect; **verify with a live fetch (Step 3) before relying on it**. MEDIUM risk (display name vs path confusion; only change path, never name).

What **MUST** be operator-approved (§6.T.3 / protected-branch boundary)

1. **GATE 1** — the Step-0 remote-set normalization (removes upstream/github remotes; rewrites origin push URLs). Confirm before proceeding.
2. **GATE 2** — disposition of the duplicate `HelixDevelopment/*` repos (rename/archive/leave). §11.4.29 applies if they are live own-org repos.
3. **Any glab api --method PUT** path-rename fallback — a write API call; needs explicit approval.
4. **helixqa main merge (Step 6)** — touches a protected branch on an own-org repo via PR. Normal PR, NO force-push; still operator-approved per §6.T.3 because it changes a protected branch.
5. **Panoptic GitHub rename** — confirm panoptic is in-scope for §11.4.29 (it is an own-org `vasic-digital` repo; GitLab side is already lowercase).

Non-risks (explicitly verified)

- **No go.sum churn** — module identifiers unchanged. No `go mod tidy` needed for the rename itself (Step 4's path fix does not change checksums either).
- **No Go import-path edits** — zero `github.com/owner/Repo` import references.
- **No parent CI/script edits** — `scripts/`, `Upstreams/`, `tools/` name none of these repos.
- **No force-push anywhere** — all operations are renames, URL rewrites, normal commits, and a normal PR. §6.T.3 satisfied.

Recommended execution order (summary)

Step 0 (backup + normalize, GATE 1) → Step 1 (rename GitHub+GitLab, GATE 2) → Step 2 (submodule remote URLs) → Step 3 (`.gitmodules` + sync + verify) → Step 4 (`llm_orchestrator` path fix) → Step 5 (submodule-internal doc/upstream updates) → Step 6 (helixqa main realignment, GATE for protected branch) → Step 7 (single Lava-parent commit + 2-mirror push).